

Step 1 — HAR vs. Wavelet-LGB  
(DM two-tailed; ↑ = wavelet wins)

|            | $h = 1$              | $h = 5$             | $h = 10$             |
|------------|----------------------|---------------------|----------------------|
| DJIA       | ↓ ***<br>$p = 0.000$ | ↓ *<br>$p = 0.018$  | ↑ ns<br>$p = 0.128$  |
| Nasdaq-100 | ↓ ***<br>$p = 0.000$ | ↑ ns<br>$p = 0.191$ | ↑ ***<br>$p = 0.001$ |
| S&P 500    | ↓ **<br>$p = 0.008$  | ↓ ns<br>$p = 0.296$ | ↑ ns<br>$p = 0.440$  |

Step 2 — Wavelet-LGB vs. Spillover-LGB  
(CW one-tailed nested; ↑ = spillover wins)

|            | $h = 1$             | $h = 5$             | $h = 10$             |
|------------|---------------------|---------------------|----------------------|
| DJIA       | ↑ *<br>$p = 0.040$  | ↑ **<br>$p = 0.004$ | ↓ .<br>$p = 0.082$   |
| Nasdaq-100 | ↑ ns<br>$p = 0.385$ | ↓ ns<br>$p = 0.936$ | ↓ ns<br>$p = 0.164$  |
| S&P 500    | ↑ *<br>$p = 0.019$  | ↓ *<br>$p = 0.026$  | ↑ ***<br>$p = 0.000$ |